COVID-19 Global Impact Analysis

Trends across waves, countries and vaccination rollouts

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1. Executive Summary

This report analyses the publicly available Our World in Data (OWID) COVID-19 dataset to understand how the pandemic unfolded across countries and time. The full analytics pipeline was followed: cleaning messy time-series data, handling missing values, deriving new metrics (7-day rolling averages, case fatality rate, doubling time) and communicating the findings visually. The report is organised around one focused question: **did faster vaccination rollout correlate with lower death rates?**

Headline findings

- **Per-capita comparison changes the ranking.** Peru (6,601 deaths per million), Bulgaria (5,670) and North Macedonia (5,422) were hit hardest.
- **Cases and deaths decoupled.** Peak case rates in the Omicron period (2022) were many times higher than earlier peaks, yet daily death rates peaked much earlier, in 2020-21.
- **Case fatality rates varied widely** among the six countries compared, from 0.46% (Germany) to 2.52% (South Africa).
- **Vaccination and deaths showed a positive correlation** (Spearman rho = 0.63, 154 countries). This reflects confounding by age, wealth and reporting quality, not a harmful effect of vaccines.

2. Objective and Data

Objective: practise the complete analytics workflow on messy real-world time-series data and communicate results to a non-technical audience.

Source: Our World in Data COVID-19 dataset (github.com/owid/covid-19-data). The file used contains 429,435 daily records for 255 locations between 1 January 2020 and 14 August 2024. Fifteen relevant columns were retained: country, continent, date, population, total and new cases and deaths (raw and per million), total vaccinations and the share of people fully vaccinated.

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy) in a Jupyter notebook.

3. Data Cleaning and Documented Assumptions

#	Issue found	Action taken and rationale
A1	Rows for World, continents and income groups (iso_code starting with OWID_)	Removed. They are aggregates and would double-count countries.
A2	Very small countries and territories	Kept only countries with population of at least 1 million, because per-million rates are unstable for tiny populations.
A3	Negative daily cases or deaths	Set to 0. Negative values are retrospective data corrections, not real negative counts.
A4	Missing daily values	Filled with 0, assuming no report was published that day.

#	Issue found	Action taken and rationale
A5	Missing cumulative values (totals, vaccinations)	Forward-filled within each country, remaining gaps (before first case) set to 0, as cumulative values cannot fall.

After cleaning, the dataset contains 266,897 rows covering 159 countries.

4. Derived Metrics

Metric	Definition	Why it is useful
7-day rolling average	Mean of the last 7 days of new cases or deaths per million	Removes weekend reporting dips so waves are easier to read.
Case fatality rate (CFR)	Total deaths / total cases x 100 (only after 100 cases)	Rough indicator of deadliness among confirmed cases.
Doubling time	7 x ln(2) / ln(cases today / cases 7 days ago)	Shows how fast an outbreak was growing; short = fast growth.
Per-million values	Counts divided by population in millions	Makes countries of different size fairly comparable.

5. Analysis of Waves and Country Comparison

Six countries from different regions were compared: India, the United States, Brazil, the United Kingdom, Germany and South Africa.

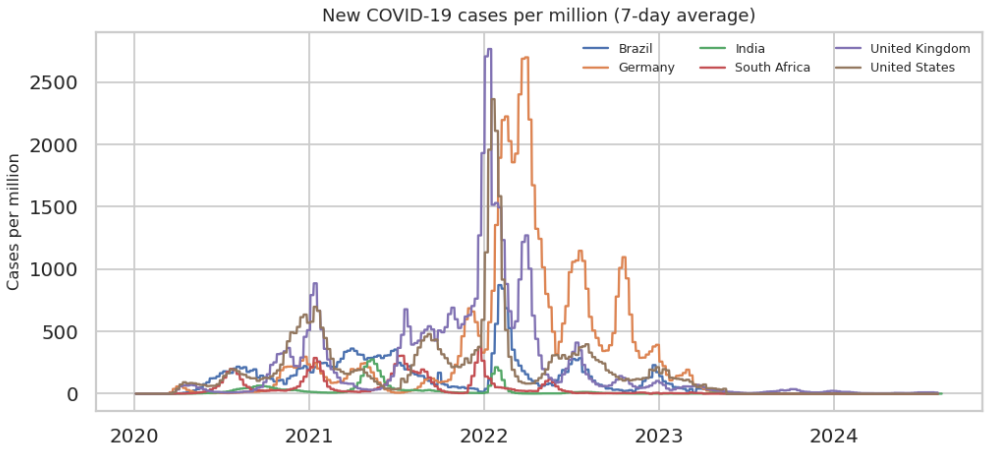


Figure 1. New cases per million, 7-day average.

Cases peaked at different times. India's largest wave came in May 2021 (Delta). South Africa peaked in December 2021, the first major Omicron wave. The United States, United Kingdom, Brazil and Germany peaked between January and March 2022, and the 2022 peaks in the UK (about 2,767 cases per million per day) and Germany (about 2,699) dwarfed anything seen earlier in the pandemic.

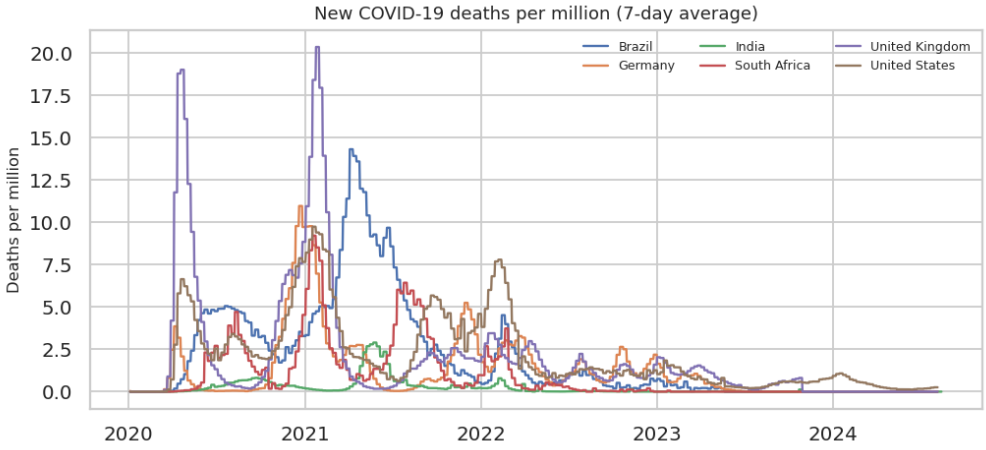


Figure 2. New deaths per million, 7-day average.

Deaths tell a different story. Death rates peaked mostly between December 2020 and April 2021, for example the UK at about 20.4 deaths per million per day in January 2021 and Brazil at about 14.3 in April 2021. Despite record case numbers in 2022, daily deaths per million stayed well below these earlier peaks.

Cumulative impact per million people

Country	Total cases per million	Total deaths per million	Case fatality rate
Germany	457,123	2,081	0.46%
United Kingdom	366,308	3,404	0.93%
United States	302,860	3,494	1.15%
India	31,599	374	1.18%
Brazil	178,368	3,339	1.87%
South Africa	65,291	1,645	2.52%

Countries with the highest cumulative deaths per million were Peru (6,601), Bulgaria (5,670), North Macedonia (5,422), Bosnia and Herzegovina (5,115) and Hungary (5,065). India's low per-million figure should be read with caution, as official death counts in several countries are widely believed to understate the true toll.

6. Vaccination Rollout and Death Rates

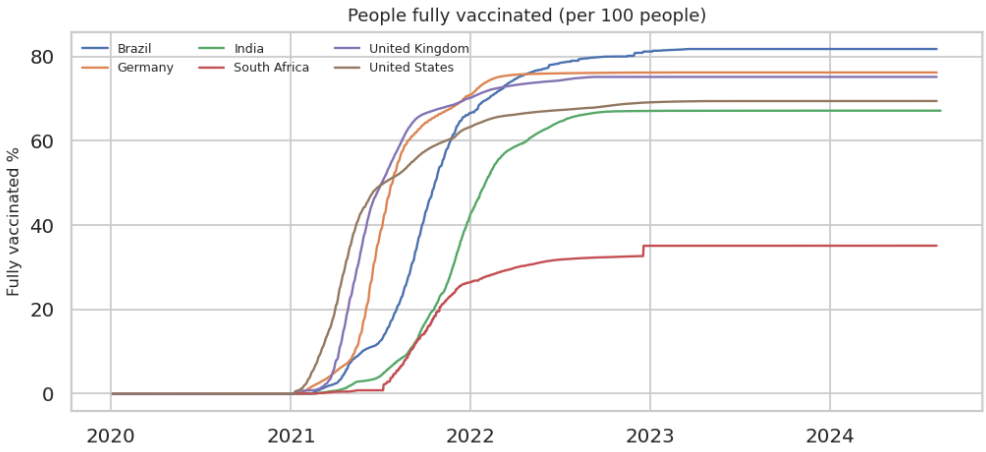


Figure 3. Share of the population fully vaccinated.

Focused question: did countries with faster vaccination by the end of 2021 record fewer deaths in 2022? Each country's vaccination coverage on 31 December 2021 was compared with its total deaths per million during 2022 (154 countries).

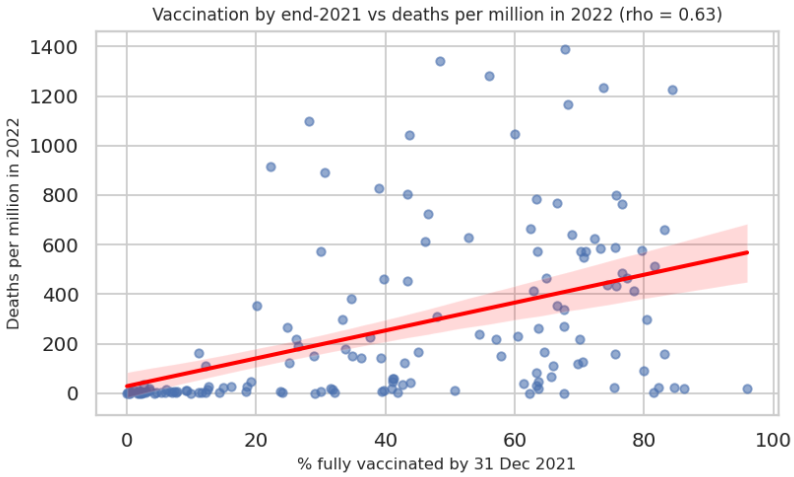


Figure 4. Vaccination coverage at end-2021 vs deaths per million in 2022. Red line shows the fitted trend.

The correlation is **positive** (Spearman rho = 0.63, p < 0.001), meaning countries with higher end-2021 vaccination coverage reported *more* deaths per million in 2022. This should not be read as vaccines causing deaths. Early-vaccinating countries tend to be wealthier, have older populations and test and report deaths more completely, while many low-vaccination countries have young populations and limited reporting, which produces the cluster of near-zero death rates on the left of the chart. In addition, the large Omicron wave of 2022 spread widely in highly connected, highly vaccinated countries.

7. Key Insights

1. Per-capita normalisation is essential. Ranking by deaths per million shows Peru, Bulgaria, North Macedonia, Bosnia and Herzegovina and Hungary as the worst affected, a very different list from raw totals.

2. Cases and deaths decoupled after 2021. Omicron-era case rates were several times higher than earlier peaks, yet death rates peaked in 2020-21. Immunity from vaccination and prior infection, and a less severe variant, likely weakened the link between infection and death.

3. Death rates among confirmed cases differed widely. CFR ranged from 0.46% in Germany to 2.52% in South Africa. Differences in testing coverage, age structure and healthcare capacity explain much of this gap, so CFR should not be used as a pure measure of virus deadliness.

4. Simple vaccination-versus-deaths comparisons are misleading. A positive correlation of 0.63 appeared across 154 countries, driven by confounding factors such as age, income and reporting quality.

5. Context matters more than any single number. Each country's outcome reflects timing of variants, demographics, health systems and data quality together.

8. Limitations

- **Correlation is not causation.** Age structure, healthcare capacity, testing and reporting quality were not controlled for.
- **Under-reporting.** Official case and death counts vary in completeness between countries and over time.
- **Single-year window.** The 2022 comparison excludes deaths in 2020-21, before and during early vaccine rollout.
- **Imputation.** Filling missing daily values with 0 may slightly understate activity in countries that reported irregularly.

9. Recommendations and Next Steps

- Add **median age** and other demographic columns from the full OWID dataset and repeat the correlation after controlling for age.
- Compare deaths **before and after** vaccine rollout within the same country instead of across countries.
- Use excess-mortality data to reduce the effect of under-reporting.
- Build an interactive dashboard (Power BI or Tableau) with country and date slicers so viewers can compare countries themselves.

10. Conclusion

The analysis shows that the pandemic's impact differed greatly by country and by time, and that the relationship between infections and deaths changed after 2021. The unexpected positive correlation between early vaccination and 2022 deaths is a useful reminder that simple cross-country comparisons can be misleading; understanding COVID-19 outcomes requires adjusting for age, healthcare capacity and reporting differences.

Data source: Our World in Data, COVID-19 dataset (data up to 14 August 2024). Analysis performed in Python.